

PARTICULAR SPECIFICATION P12

## Operations and Maintenance (OM)

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## P12.1 OVERVIEW

- (a) Following substantial completion of the whole of the Works, the Contractor will be required to provide Operations and Maintenance (OM) services. OM services will be administered under the Goods & Services Conditions of Contract (GSCOC).
- (b) The OM services will need to be provided during on-going construction of the TWRP facilities by Other contractors. In providing the OM Services, the Contractor will need to cooperate fully with the Board and co-ordinate OM Services with the Other contractors.
- (c) This section describes the minimum requirements for the Operations and Maintenance (OM) Period of the Contract. This Specifications shall be read in conjunction with other sections of the Contract.
- (d) For the avoidance of doubt, building services and environmental controls which includes general area cleaning (including toilets, testing room cleaning, waste management) and housekeeping shall be included in the scope of work during the OM period. The Contractor shall work with the Board appointed Pest Control agent for access and management of the Site.
- (e) OM period is shown in Table P12-1.

**Table P12-1: OM Period**

| OM Period | Start of OM Period                          | Nominal Duration (months) | Facilities                                                                                                                       |
|-----------|---------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| OM3       | Upon issuance of CSC for whole of the Works | 12                        | OM of Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities and ancillaries required for the operation of the plant |

- (f) The Contractor shall provide all activities and spare parts for routine service, preventive and breakdown maintenance for the plant and equipment including all special tools, instrumentation and analysers.
- (g) The Contractor shall ensure the support of the manufacturer and where necessary, engage the manufacturer's representatives to carry out specialist servicing and maintenance of equipment.
- (h) The following will be provided by Other contractors during OM3:
  - (i) Electricity;
  - (ii) Water; and
  - (iii) Supply of Chemicals.

- (i) The Contractor shall include all other costs for operations and maintenance of the facilities during OM3. This will include but not limited to the following operations and maintenance costs:
  - (i) Labour (e.g. including labour for topping up and unloading of chemicals);
  - (ii) Maintenance and asset replacement;
  - (iii) Grease and lubricating oil;
  - (iv) Diesel for generators;
  - (v) Instrumentation consumables;
  - (vi) Sampling and analysis for investigation purpose and as specified in Appendix P12A;
  - (vii) Replenishment of seed sludge for Sidestream Nitrogen Removal in the event of loss of Anammox sludge in the Reactors due to operational failures (If Cross-seeding is not possible); and
  - (viii) Annual certification for all statutory testing and certification (for example lifting equipment, fire extinguishers, fire protection systems, and the like).
- (j) If the Board considers that the Contractor is either not providing the required operations and/or maintenance services, the Board will have the right to appoint another company to carry out the operations and/or maintenance and deduct all incurred costs from the Contract Sum.

## **P12.2 OVERVIEW OF SCOPE OF WORK DURING EACH IO AND OM PERIOD**

- (a) The Scope of Work to be carried out during each IO and OM period shall include but shall not be limited to:
  - (i) Operation of plant and equipment, 24 hours per day, 7 days per week (24/7) as directed by the Board;
  - (ii) Maintenance (routine and preventive) of plant and equipment during normal working hours;
  - (iii) Rectification of breakdowns of plant and equipment within the specified time periods;
  - (iv) Reporting regularly on the operation, performance and serviceability of plant and equipment;
  - (v) Updating of Board's Asset Management System (AMS);

- (vi) Site security 24 hours per day, 7 days per week (24/7); and
- (vii) Develop and implement an Operations and Maintenance Plan (OMP) that captures all associated activities during the IO and OM periods.

### **P12.3 OPERATIONS AND MAINTENANCE PLAN**

#### **P12.3.1 Operations and Maintenance Plan (OMP)**

- (a) The OMP shall refer to equipment manufacturers' O&M instructions, statutory requirements, process control narratives, etc. The OMP shall be submitted to the Board at least ninety (90) days prior to the date for the commencement of each IO and OM period. Comments from the Board on the OMP shall be addressed by the Contractor prior to the start each IO and OM period. The OMP shall describe how the operations and maintenance of the plant and equipment will be undertaken during the IO and OM period and shall cover as a minimum:
  - (i) Normal operation;
  - (ii) Start-up;
  - (iii) Shutdown;
  - (iv) Low flow operation;
  - (v) Peak flow operation;
  - (vi) Emergency Response Plan;
  - (vii) Incident Response Plan;
  - (viii) System control and design target requirements;
  - (ix) Control optimisation strategy;
  - (x) Toxic influent management;
  - (xi) Alarm management and response times to be implemented;
  - (xii) Preventive Maintenance Plan and Replacement Plan;
  - (xiii) Sampling and Monitoring Plan;
  - (xiv) Data Management and Reporting; and
  - (xv) Operations and Maintenance Handover Management Plan.

#### P12.4 DEFECTS LIABILITY PERIOD (DLP)

- (a) The Contractor shall note that the Defects Liability Period (DLP) for Part 1 will be simultaneous with the Part 2, OM period. The DLP for Part 1 will be administered under the PSSCOC.
- (b) The Contractor's responsibilities during the Part 1 DLP are broadly defined in the PSSCOC. This includes finding and rectifying defects, and the provision of all labour, parts and materials for the rectification of such defects at no additional cost to the Board. The Contractor shall note that the rectification of defects will not be considered as maintenance during the IO and OM periods.

#### P12.5 SAMPLING AND ANALYSIS DURING THE IO AND OM PERIOD

- (a) The Contractor shall provide the sampling and analysis for investigation purposes and as specified in Appendix P12A as a minimum, during the IO and OM periods. All costs associated with sampling, and analysis, including but not limited to equipment, storage, transportation, and labour shall be included in the Contract Sum.
- (b) The methods to be used to analyse the listed parameters shall be as given in Standard Methods for the Examination of Water and Wastewater; 21st Edition, 2005; published by APHA, AWWA and WEF.

#### P12.6 OPERATION OF PLANT AND EQUIPMENT DURING IO AND OM PERIODS

- (a) During each IO and OM period, the Contractor shall as a minimum:
  - (i) Work with the Board to operate the plant and equipment as intended to meet the needs of the TWRP Project, including interfaces with IWMF;
  - (ii) Take the lead in managing the safe and proper operation of plant and equipment;
  - (iii) Work with the Board, other TWRP interfacing contractors and IWMF to facilitate the testing and commissioning of other TWRP contracts;
  - (iv) Record the plant flow data, performance parameters and laboratory results in the format specified by the Board.
  - (v) The Contractor shall note that TWRP laboratory may not be available to the Contractor during IO and OM period. The Contractor may set up a testing area at suitable locations as agreed by the Board. The testing area shall be equipped with the following but not limited to:
    - 1) Temporary water supply and wash basin for washing of laboratory glassware; and
    - 2) Suitable waste container for disposing samples with spent reagents. The waste shall be properly disposed in compliance with Environment, Health and Safety requirements;

- (vi) Update the operating and maintenance information based on experience gained in operating the plant during the period;
- (vii) Fine tune and adequate adjustment control of the plant and equipment to optimise reliability, performance and energy efficiency;
- (viii) Ensure safe operation of the plant and equipment, for seven days per week, 24 hours per day with fulltime and adequate onsite staffing. Duty Operations and Maintenance Manager to be contactable by the Board at all times;
- (ix) Control and manage security and access to the Digestion Pre-treatment and Sidestream Nitrogen Facilities, 24 hours per day, 7 days per week (24/7). This includes but not limited to the following, subject to Board approval:
  - 1) Co-ordinate with the Board appointed site security agent to ensure the Site is secure;
  - 2) Control site access and authorisation; and
  - 3) Rooms containing equipment are locked at all times and have controlled access;
- (x) Monitor and respond appropriately to all alarms within set time frames and to maintain the operational performance of the plant, including out-of-hour times;
- (xi) Ensure that all consumables are provided for the correct operation of all plant and equipment at all times;
- (xii) Co-ordinate with the Board in a timely manner for the supply of chemicals for the plant operation to ensure a continuous supply of chemical;
- (xiii) Co-ordinate with the Board on the topping up and unloading of chemicals ensure that the activities are carried out safely;
- (xiv) Develop and implement Emergency Response Management procedures for implementation during the IO and OM periods including emergency situations such as fire, power outage and loss of communication systems at the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities;
- (xv) During Part 1, as part of testing and commissioning, the Contractor shall provide training of the Board's personnel as specified. The Contractor shall provide two (2) additional on-the-job training sessions for the Board's personnel as directed by the Board during the OM period. The Board will provide 1 month's advance notice for the sessions which shall be held over a continuous two (2) week period, for up to 20 staff at one time, and shall include but not be limited to:

- 1) Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities operation training; and
  - 2) Equipment training.
- (xvi) The intent of these training sessions is for the Contractor to pass on operational experience to the Board's personnel. Training shall be carried out by the Contractor's operations and maintenance personnel. Input from vendors is not required.
- (xvii) Adopt the Board's Asset Management described in Particular Specification P13 Adoption of the Board's Asset Management System. The Contractor shall maintain and update the asset management systems;
- (xviii) Provide resourcing, roster and organisational charts to operate and maintain the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities, including:
- 1) Staffing organisation including key personnel;
  - 2) Shift schedules and call out rosters;
  - 3) Staff responsibilities; and
  - 4) Staff qualification;
- (xix) Ensure meetings and attendance requirements are met;
- (xx) Ensure that the Digestion Pre-treatment and Sidestream Nitrogen Removal are kept in continuous operation (24/7), as specified to meet the plant performance requirements during the IO and OM periods.
- (xxi) The Contractor shall coordinate with the Board in a timely manner for all equipment, consumables and spares stored in the permanent on site TWRP storage facilities (by Others) during the IO and OM periods, to be transport to the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities as required by the Contractor. Transportation of stored equipment, consumables and spares shall be provided by the Contractor from the permanent on site TWRP storage facilities to Digestion Pre-treatment or Sidestream Nitrogen Removal Facilities as required.

## **P12.7 MAINTENANCE OF PLANT AND EQUIPMENT DURING EACH IO AND OM PERIOD**

### **P12.7.1 Scope of Maintenance Services**

- (a) The maintenance services provided shall cover all the Plant and equipment, included in each IO and OM period as well as the environmental controls as specified in Part 1 and approved OMP.

- (b) During OM period, the Contractor shall note that the Board will engage its own maintenance contractors for the maintenance of Building Services in Digestion Pre-treatment Sidestream Nitrogen Removal Facilities. The Contractor shall cooperate fully with the Board to facilitate such maintenance activities of Other contractors during the OM period.
- (c) Maintenance includes but is not necessarily be limited to engaging all management and maintenance personnel, all equipment servicing requirements, routine 'preventive' maintenance requirements as well as 'breakdown' maintenance. The work shall also include for the provision of all spare parts whenever required, as well as the provision of all lubricants, fittings, gaskets, fixings and fasteners and any other consumable items associated with all maintenance activities.
- (d) "Preventive Maintenance" shall be performed at least every month or as otherwise required during each IO and OM period. It is the responsibility of the Contractor to provide adequate manpower with relevant expertise to carry out the work.
- (e) All labour costs, parts and consumables cost incurred in the preventive maintenance and breakdown/corrective maintenance as well as modification works, menial and miscellaneous works shall be deemed to have been included in the Contract Sum. The Contractor shall be responsible for ensuring that sufficient numbers of trained personnel are available to perform maintenance.
- (f) At the end of the OM period, the Contractor shall furnish to the Board a full set of all special tools, instruments and safety equipment necessary for all maintenance activities as a minimum as was required during the IO and OM period.
- (g) All daily, weekly and monthly checks, calibrations or adjustments that are considered part of the operations and maintenance requirements as defined under the Contractors developed Standard Operating Procedures (SOP) and equipment manufacturer's recommendations shall be provided.
- (h) The maintenance services provided shall include for:
  - (i) All issues deemed to be the result of equipment, instrument, control logic, hardware, software and communications errors or failures;
  - (ii) All maintenance or repair activities, including any transportation or other expenses incurred;
  - (iii) Any maintenance or repair that is required due to damage caused through incorrect installation or maintenance of equipment;
  - (iv) Any maintenance or repair that is required due to damage resulting from inadequate testing or commissioning;
  - (v) Any maintenance or repair that is required due to any foreign matter, debris or anything within the process piping system that causes process or equipment impairment;

- (vi) Any maintenance or repair that is the result of defective equipment;
- (vii) Any plant, equipment, tools, lighting, safety equipment and crane or other lifting equipment needed for any maintenance activity including equipment removal and replacement;
- (viii) Any measuring instruments, calibration equipment, laboratory equipment or any other special equipment, test and specialist personnel needed for any maintenance or testing activity;
- (ix) All scheduled Preventive Maintenance checks, calibrations or adjustments that are considered part of the operations and maintenance requirements for all the mechanical and electrical equipment, instruments and controls, not limited to service checks (e.g. daily, weekly, monthly, etc.), adjustments and all other activities that are considered part of the operations and maintenance requirements as defined under the Standard Operating Procedures (SOPs) and equipment manufacturer's manuals;
- (x) All other routine preventive and breakdown/corrective maintenance as well as equipment service checks and activities;
- (xi) All consumables;
- (xii) All maintenance or repairs that are deemed to be the result of:
  - 1) Equipment, instrument, control logic, software, hardware or communications errors or failures;
  - 2) Equipment wear and tear;
  - 3) Operational defects and abnormalities in the process piping system that causes equipment damage.
- (i) Any plant, equipment and tools required for any maintenance activity inclusive of statutory inspections, tests/certifications, removal and replacement of the plant, equipment and tools.
- (j) Replacement of any plant or equipment (e.g. valve/liner) as and when required due to failure (breakdown/corrective maintenance) as per requirement stated in the Specifications.
- (k) Any maintenance or repair to any temporary storage or containment of chemicals, including any associated piping, auxiliaries and fittings.
- (l) Any menial and miscellaneous works due to preventive maintenance and breakdown/corrective maintenance activities. The nature of work shall include, but not limited to, the following:
  - (i) Clear chokes/blockages in pipes, sampling tubes, etc.;

- (ii) Clean and clear any spillage from any plant and equipment;
  - (iii) Cleaning of the facilities;
  - (iv) Clear debris due to scheduled maintenance, urgent repair or any modification works carried out;
  - (v) Touch up of plant and equipment due to any preventive, urgent/ breakdown/ corrective work; and
  - (vi) Removal of rust on plant and equipment, repair of surfaces and repainting.
- (m) The Contractor shall provide to the Board the proper documentation for all activities carried out, which may include as a minimum:
  - (i) Monthly and annual maintenance reports to meet the requirements of General Specification G1.2 Information Management;
  - (ii) Records for routine and preventive maintenance checklists;
  - (iii) Diagnostic reports for breakdown/corrective maintenance. This report shall include all parts or materials found faulty and replaced; and
  - (iv) Any other reports not mentioned herein but reasonably requested by the Board.
- (n) The Contractor will be responsible for housekeeping and cleaning the area where any preventive or breakdown maintenance has been performed. It is also the Contractor's responsibility for disposal of all waste materials generated while performing such activities, and to present the disposal records to the Board if requested. The Contractor shall obtain a movement slip from the Board before any disposal works is carried out.
- (o) The Contractor shall provide to the Board detailed Maintenance Reports for all routine, preventive maintenance and breakdown maintenance activities. All daily and weekly activities, including all equipment and system checks shall be fully detailed and reported in a consolidated Monthly Report. The Monthly report shall be submitted within one (1) week after the end of each month. Payment shall only be made after Monthly Reports have been satisfactorily submitted in full by the Contractor and approved by the Board.
- (p) All repairs and replacements shall be made to a specified standard and in a timely manner and must not be deferred due to cost pressures or other factors. At the end of the IO and OM periods, the Plant must be in a sound, proper and well-maintained condition – there must not be the necessity for a major overhaul if the decision is taken to continue operation into the future.

- (q) The Contractor shall provide a team of qualified, experienced management and skilled maintenance personnel for the effective management, co-ordination and execution of the maintenance programme and all associated maintenance tasks.
- (r) All maintenance personnel deployed by the Contractor shall be familiarised with the operation of the Plant.
- (s) The Contractor shall be responsible for providing all necessary management and resources and for providing all necessary maintenance functions to support the operations of the Plant to ensure compliance with the Plant performance requirements.

#### **P12.7.2 Preventive Maintenance**

- (a) Preventive maintenance and equipment servicing shall be carried out at intervals specified by the relevant equipment manufacturers and the Contractor shall submit a Preventive Maintenance Plan to the Board for approval as part of the OM Plan.
- (b) The maintenance activities to be included in the Preventive Maintenance Plan shall be compatible with the details provided in the operations and maintenance information for the Plant. The maintenance activities shall include all daily, weekly, monthly and annual activities required to ensure that all the equipment will be serviced and maintained properly to maximise its life-cycle and minimise breakdown or failure.
- (c) Any activities not described in the Contract's operations and maintenance information but adopted during the IO and OM periods shall be included in the operations and maintenance information and re-issued at the Contractor's cost.
- (d) These maintenance activities shall also include the routine service, maintenance and calibration of all instruments and analysers, and specialised forms of on-line instrumentation installed in the Plant.
- (e) The maintenance of any specialised equipment and instrumentation systems, analysers and control systems shall only be carried out by qualified and certified manufacturer's representatives. The Contractor shall be deemed to have included for all service agreements and costs in the prices during the IO and OM periods.
- (f) Failure on the part of the Contractor to carry out preventive maintenance in accordance with the maintenance schedule and the manufacturer's recommendations may result in a deduction to the Contractor's monthly payment as determined by the Board.

**P12.7.3 Breakdowns / Corrective Maintenance**

- (a) The Contractor shall perform plant troubleshooting and rectification if any abnormality or breakdown of equipment is detected, including items that may fall beyond the normal routine maintenance and preventive maintenance schedules. The Contractor shall identify the root cause of any process upset. If any fault recurs two or more times in a period of one (1) year (or over any shorter critical period determined by the Board), the Contractor shall engage the equipment manufacturer if instructed by the Board for the identification and documentation of the root cause. The Contractor shall bear the costs of any equipment overhaul whether due to breakdown or per recommendation of the equipment manufacturer. Documentation of the repairs are to be provided in a format approved by the Board.
- (b) Attending to breakdown/corrective maintenance, ad-hoc work and additional works shall not relieve the Contractor from carrying out the scheduled preventive maintenance works. Hence, the Contractor shall provide additional manpower to attend to breakdown/corrective/urgent repair works, ad-hoc work and additional works if the normal rostered full-time staff is insufficient.
- (c) The Contractor shall ensure that suitably qualified maintenance personnel, in sufficient numbers, are available to attend to any breakdown/corrective maintenance situation in less than 12 hours.
- (d) Failure to attend Site within 12 hours will result in the Contractor incurring an administrative charge (AC) of \$100 per hour, for every hour of delay. The Board's decision on whether the Contractor has failed to respond within the stipulated timeframe shall be final and binding.
- (e) Failure to replace or repair the equipment or system and fully recommission and return to normal operation within five (5) days by default, will result in the Contractor incurring an administrative charge (AC) of \$100 per hour for every hour of delay. The Board's decision on whether the Contractor has failed to respond within the stipulated timeframe shall be final and binding.
- (f) The Contractor shall submit a breakdown/ corrective maintenance report to the Board within seven (7) working days from rectification. The format of the report shall be agreed with Board. The report shall include documentation/photographic evidence of the occurrence of the breakdown/ corrective maintenance event and rectification of the situation.
- (g) A "critical breakdown / corrective maintenance" is defined as the failure or breakdown of any item of equipment or system that could directly or indirectly affect the ability of the Plant to perform as specified. The Board's decision on what does and does not constitute a critical breakdown shall be final and binding. The following are examples of situations that are considered a critical breakdown and which must be addressed without delay:
  - (i) Inability of the Plant to reliably receive, treat and discharge the specified quantity and quality of sludge or centrates;

- (ii) Inability of the Works to reliably and continuously receive the specified quantity and quality of influent from any other sources as specified; and
- (iii) Inability of the Works to reliably and continuously meet the performance requirements as specified.
- (h) For any possible critical breakdown maintenance, the Contractor's personnel shall be responsible for notifying the Board promptly, within thirty (30) minutes of any breakdown/corrective maintenance situation. The Board will issue a notice of critical breakdown.
- (i) The Contractor and the Board shall immediately convene a discussion to assess the impact of the breakdown/corrective maintenance situation. The Board's decision on what does and does not constitute a critical breakdown shall be final and binding.
- (j) Formal notice shall be issued to the Contractor of the nature of the breakdown and any actions required during the critical breakdown period;
- (k) Once the situation has been rectified, the Contractor shall issue a formal notice to the Board to notify of the rectification of the situation.
- (l) The Contractor shall submit a critical breakdown/corrective maintenance report to the Board within three (3) working days from rectification. The format of the report shall be agreed with the Board. The report shall include documentation/photographic evidence of the occurrence of the breakdown/corrective maintenance event and rectification of the situation.

**P12.7.4 Required Contractor Reaction to Critical Breakdown during IO and OM Period**

- (a) For critical breakdown/corrective maintenance events, the Contractor shall carry out the following:
  - (i) The Contractor shall deploy additional manpower and/or equipment to manage the situation as necessary;
  - (ii) The Contractor shall activate the applicable response plan for the situation;
  - (iii) The Contractor shall carry out the rectification works immediately and complete the rectification works expeditiously;
  - (iv) Where a critical breakdown/corrective maintenance is caused by an electrical fault or equipment, the Contractor's Licensed Electrical Worker (LEW) of the appropriate grade and his Electrician shall be present to rectify the fault; and
  - (v) Where the critical failure involves the loss of Anammox sludge activity in the Reactors, the Contractor's process specialist shall be present to recover the operation.

- (b) The Contractor shall ensure that suitably qualified maintenance personnel, in sufficient numbers, are available to attend to any critical breakdown/corrective maintenance situation in less than three (3) hours.
- (c) Failure to attend Site within three (3) hours will result in the Contractor incurring an administrative charge (AC) of \$200 per hour, for every hour of delay. The Board's decision on whether the Contractor has failed to respond within the stipulated timeframe shall be final and binding.
- (d) Failure to replace or repair the equipment or system and fully recommission and return to normal operation within twenty-four (24) hours by default, or a period as approved by the Board, will result in the Contractor incurring an administrative charge (AC) of \$1000 per hour for every hour of delay. The Board's decision on whether the Contractor has failed to respond within the stipulated timeframe shall be final and binding.
- (e) Refer to P12.8.1 for details of administrative charges (AC).

#### **P12.8 MONITORING AND REPORTING**

- (a) For the entire duration of each IO and OM period, the operations and performance of plant and equipment shall be monitored by the Contractor and reported on as described herein.
- (b) The Contractor shall develop a Data Management and Reporting Plan as part of the OM Plan.
- (c) The Data Management and Reporting activities shall, at a minimum, address:
  - (i) The Board's data requirements including any BIM requirements, Asset Management System requirements, proposed exchange methodology, format and frequency of data transfers;
  - (ii) The management of turnaround time for analysis performed on samples by the on-site laboratory and an external laboratory accredited by the Singapore Accreditation Council (SAC) for laboratory testing and product certification;
  - (iii) Auditing process conducted and data transfer to monthly reporting for the Board's approval;
  - (iv) Uploading of the on-site and off-site analytical results into an approved Board's data management system including security protocols, format and frequency;
  - (v) The reporting format of the on-site and off-site test results;
  - (vi) The reporting format of the Plant flow data;
  - (vii) Production of daily, weekly and monthly operational performance reports to the key people who receive these reports;

- (viii) Production and distribution of regulatory required reports such as safety, environmental, spills or overflows, etc for both monthly and annually;
  - (ix) Production and distribution of internal and external stakeholder reports as required; and
  - (x) Maintenance information, including maintenance history, backlog and calibration records.
- (d) The Contractor shall provide and use the data management system to generate the required reports. The daily, weekly, monthly and annual reports, as a minimum, shall include:
- (i) Regulatory reports as required;
  - (ii) Internal reports as requested by the Board and, as minimum, to include;
  - (iii) Optimisation of the plant and equipment;
  - (iv) Monthly status and summary of performance; and
  - (v) Process specification issues or activities.
- (e) The Contractor shall report weekly and monthly on the following:
- (i) Volume and load of pre-dewatered sludge processed;
  - (ii) Volume and load of centrate treated;
  - (iii) Volume of process or foul gas produced from Thermal Hydrolysis Process;
  - (iv) Volumes of steam used;
  - (v) Volumes of potable, NEWater and G2D water used;
  - (vi) Energy and chemicals consumption and Key Performance Indices required by TWRP;
  - (vii) Staffing levels and associated hours of engagement (including afterhours attendance);
  - (viii) Safety Reports - incidents, injuries, near misses;
  - (ix) Incidents of loss, theft or vandalism;
  - (x) Non-compliances;
  - (xi) Noise and odour complaints;
  - (xii) Level of effort associated with maintenance and repair procedures;

- (xiii) Calibration results of flow meters and analysers;
  - (xiv) Confirmation that preventive maintenance and repair is being performed on all equipment in accordance with manufacturers' recommendations;
  - (xv) Parts used in maintenance, including stock and non-stock items;
  - (xvi) Equipment and process downtime for critical assets and processes, asset reliability and plant availability;
  - (xvii) Corrective actions/improvements/modifications made in response to incidents that have occurred; and
  - (xviii) Other statistics, feedback or commentary that may be required from time to time and as directed by the Board.
- (f) Failure to submit any of the reports mentioned herein or updating of the data management system which includes BIM and AMS, will result in the Contractor incurring an administrative charge (AC) of \$100 per day, for every day of delay. The Board's decision on whether the Contractor has failed to respond within the stipulated timeframe shall be final and binding.

**P12.8.1 Administrative Charges for Non-Performance of Services**

- (a) In addition to any other rights and remedies under the Contract, in the event the Contractor fails to meet the performance standard or Specifications as stated in this Contract, the Contractor shall pay the Board or shall allow the Board to deduct the administrative charges ("AC") as stated in this Specifications from any sums due or shall become due to the Contractor under the Contract (if any). The Board may, at his sole discretion, engage any Other contractors or employ and pay other persons to do whatever may be necessary to give effect thereto and recover any additional cost, loss, expenses and damages incurred or suffered therewith plus an administrative fee and management fee of the total cost paid (as per Clause 49 of Conditions of Contract for Goods and Services) from the Contractor.
- (b) Provided that the Contractor is able to provide written evidence or other reasonable justifications for his failure in meeting the contract requirement, the Board may consider waiving the AC.
- (c) The cumulative AC pursuant to the Specification for failure to meet the contract requirements shall be limited to 100% of the Contract Price.
- (d) For avoidance of doubt, the AC referred in this Specification is independent from the Clause 49 Administrative Fee and Management Fee.

## P12.9 PERSONNEL REQUIRED DURING EACH IO AND OM PERIOD

- (a) The Contractor shall submit to the Board for approval before deployment, a detailed Organisation Chart for each IO and OM period, identifying all full-time key personnel (including Operations and Maintenance Manager, Supervisor, Registered WSH Coordinator and all members of the operations and maintenance team). The Contractor shall provide full details of the relevant qualifications and experience of the Operations and Maintenance Manager and all operations and maintenance personnel, including full curriculum vitae (CV). This submission shall include reference to the experience of the nominated personnel in maintaining the types of equipment installed at the Plant.
- (b) The full details and CV of the operations and maintenance team, including the shift roster, shall be submitted within four (4) weeks before each IO and OM period for the approval of the Board. Any deviation of the project team members qualifications from that specified shall be subject to the Board's discretion and approval.
- (c) Technicians shall be able to articulate, read and write in English and should also be able to use personal computer to key in data if required. The Board reserves the right to reject the deployment or remove any technician already deployed if in the opinion of the Board, he or she does not meet the requirements specified, to be deployed at the Plant.
- (d) The Contractor shall submit the names of the reserved skilled Technicians and Supervisors who have similar minimum qualifications and experience, or qualification approved by the Board for covering of duties. The reserved skilled Technician and Supervisors must be familiar with the Plant operations, maintenance and troubleshooting requirements. Any arrangement for covering duties shall be made with the prior notification (at least two weeks' notice) and approval of the Board.
- (e) The Contractor shall provide the required competent manpower to operate the plant and equipment seven (7) days a week, twenty-four (24) hours a day. As a minimum, the Contractor shall provide the minimum number of operators stated below to operate the plant and equipment. All operations personnel shall have a minimum NITEC or equivalent qualification, are familiar with the process of the Plant and have experience in Plant operation. As a minimum, the Contractor shall provide these during each IO and OM period:
  - (i) Two (2) Operators on-site for each shift to operate the plant and equipment.
  - (ii) Full time Operations and Maintenance Manager during the working hours 8.00am to 5.30pm, Monday to Saturday (excluding Public Holidays). A duty O&M Manager is to be available at all times.
- (f) The Contractor shall provide the required competent manpower required to perform the sampling and analysis as specified during each IO and OM period. This manpower shall be in addition to the full-time Operators, and the technician(s) shall be suitably qualified to conduct the testing and analysis.

- (g) The Contractor shall provide the required competent manpower to ensure all preventive and breakdown maintenance are completed as per the stipulated work completion timelines stated under the preventive and breakdown maintenance specifications. As a minimum as specified below to carry out the Preventive Maintenance and Breakdown Maintenance during working hours, the manpower below shall be provided, and the Contractor shall provide additional manpower as necessary to ensure all preventive and breakdown maintenance are completed as per the stipulated work completion timelines. The Contractor shall provide, during normal working hours:
  - (i) During each IO and OM period, at least three (3) full-time skilled technicians with not less than three years' relevant experience in maintaining equipment similar to that provided under the Contract and with a minimum NITEC or equivalent qualification. Of the three (3) skilled technicians:
    - 1) One (1) shall be a licenced electrical worker (LEW) Grade 7 or better;
    - 2) One (1) shall be in mechanical; and
    - 3) One (1) in electrical and instrumentation.
  - (ii) The full-time supervisor on-site shall have a diploma in mechanical engineering or equivalent with not less five (5) years relevant experience in mechanical equipment maintenance of a similar process plant.

#### P12.10 ENVIRONMENT, HEALTH AND SAFETY MANAGEMENT

- (a) The provisions of Particular Specification P9 Health, Safety and Environmental shall apply for OM period.

#### P12.11 QUALITY MANAGEMENT

- (a) The provisions of General Specification G2.22 Project Quality and Execution shall apply.

#### P12.12 PERFORMANCE MANAGEMENT

- (a) The Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities shall be operated, maintained and managed so that the following obligations are met at all times (i.e. on a twenty-four (24) hour, seven (7) day per week basis) during the IO and OM periods:
  - (i) The plant and equipment are operated as specified by the relevant equipment manufacturers and maintained in accordance with the maintenance schedule and the manufacturer's recommendations;
  - (ii) The performance requirements demonstrated during the Plant Performance Tests are achieved to the fullest extent; and

- (iii) The Board's internal and external data collection and reporting obligations are met.
- (b) If the amount of sludge is insufficient to allow the maximum design number of units operating during the OM Period, the Board may at his absolute discretion, agree with the Contractor the sustainable sludge flow range and the corresponding number of units to be operated. Under such circumstances, the Contractor shall work with the Board to develop an appropriate Operations and Maintenance Plan for the OM period to allow a fully functional plant.

#### P12.13 OPERATIONS AND MAINTENANCE (OM) HANDOVER MANAGEMENT

- (a) The Contractor will need to develop an OM Handover Management Plan to ensure that the Contractor transfers operational responsibilities back to the Board or its nominated operator in a manner that does not impact the operation of the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities performance or compliance. The OM Handover Management Plan will not only need to cover the actions required to physically de-mobilise the Contractor's personnel and equipment, but also those actions necessary to ensure that the Board or its nominated operator have the capacity and capability to sustainably operate and maintain the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities .
- (b) The Contractor shall provide an OM Handover Management Plan for the effective implementation for when the following events occur:
  - (i) Completion of the OM Period; or
  - (ii) Severe breaches or continuous lack of demonstrated adherence to operation especially related to Plant Performance, Operations and Maintenance, Asset Management, Environmental and Licence Compliance and Safety requirements.
- (c) The Contractor shall develop and submit an OM Handover Management Plan to the Board for review and approval as part of the OM Plan.
- (d) The Contractor shall implement the OM Handover Management activities, in consultation with the Board no less than four months prior to the planned Handover Date. On early termination by the Board, the Contractor shall implement those elements of the OM Handover Management Plan relevant to early termination.
- (e) The Contractor's OM Handover Management Plan shall, as a minimum, address:
  - (i) The Contractor shall identify the procedure in how information will be provided to the Board;
  - (ii) The Contractor shall provide procedures in the transition of appropriate staff;
  - (iii) The Contractor shall provide protocols and procedures for the induction and/or training of the Board staff in site procedures that are necessary to ensure a safe work environment and the continuity of compliant operations at handover;

- (iv) The Contractor shall handover the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities assets in a suitable condition for continued operation and if necessary, replace or repair any assets that the Board deem to be in an unacceptable condition, this activity may include a condition assessment being undertaken to determine what assets may require remediation;
- (v) The Contractor shall provide protocols and procedures for the handover of all asset and operational data, correspondence with any Authority, and all other records in a form readily useable by the Board;
- (vi) The Contractor shall provide procedures for the handover of the AMS, commensurate with actual configuration, operations, buffers, redundancies, processes, operations and the corrective, preventive maintenance and inspection regime at time of handover;
- (vii) The Contractor shall provide procedures for the handover and transfer of licences for use of any systems;
- (viii) The Contractor shall provide protocols and procedures for the transfer of spares, plant and equipment and tools (including specialist tools);
- (ix) The Contractor shall provide procedures to make the necessary arrangements for stocktake of spare parts and hot standby assets to enable reconciliation with the opening spare parts and hot standby assets levels at the start of IO and OM Period, and evidence of the applicable supply price;
- (x) The Contractor shall provide procedures for the transfer of security entitlements, passes, identification, IT user and administrator logins, or any other similar matters to enable full and effective use of all of facilities, infrastructure and operating systems; and
- (xi) The Contractor shall provide procedures for any other transition or handover arrangements the Board deems necessary to ensure seamless delivery of the Services to the Board.

#### P12.14 APPENDIX

- (a) APPENDIX P12A – Sampling and Analysis.

**END OF SECTION**